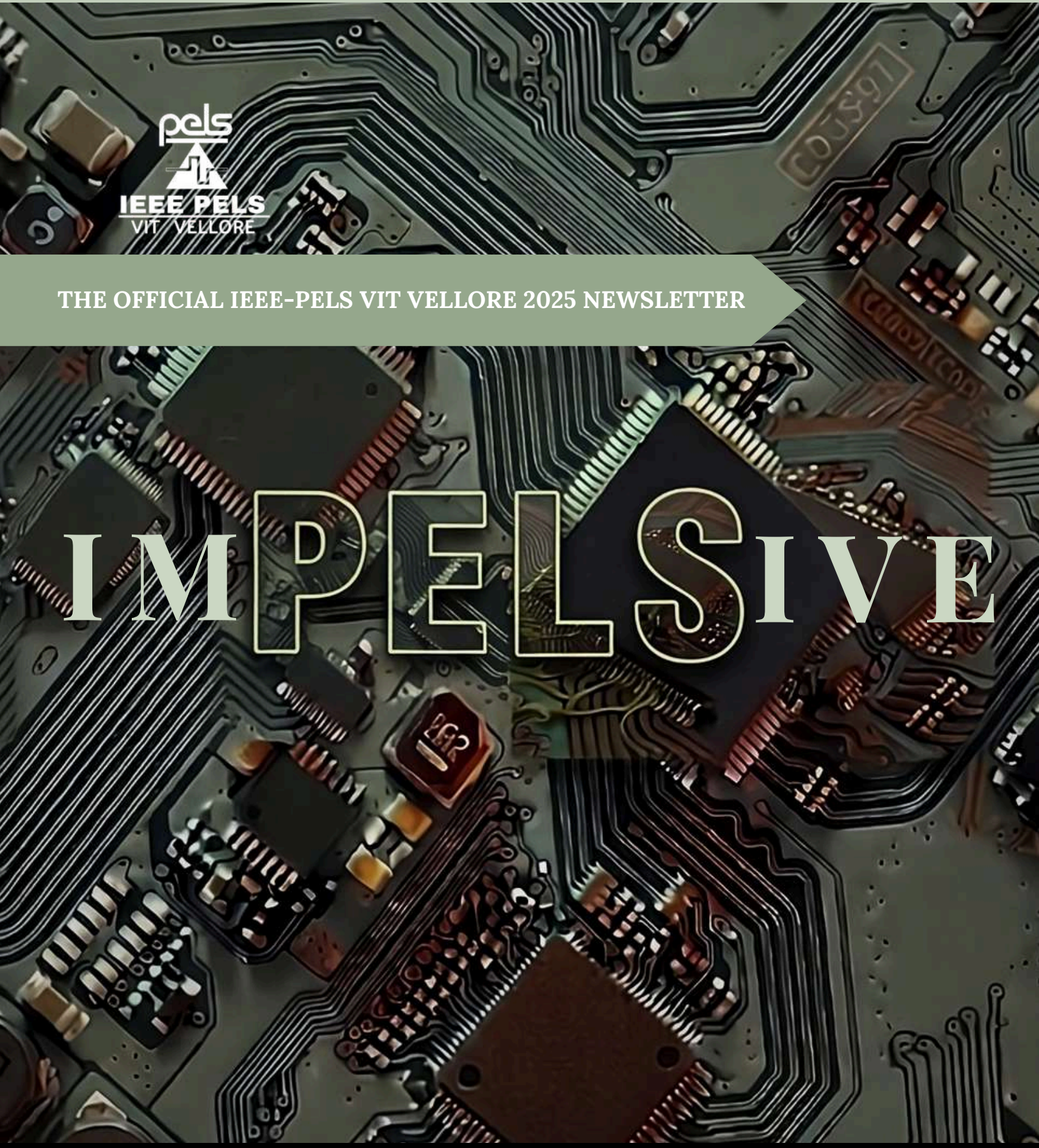
OFFICE OF STUDENT'S WELFARE



THE OFFICIAL IEEE-PELS VIT VELLORE 2025 NEWSLETTER

IMPELSIVE

ieeepels@vit.ac.in

IEEE-PELS VITV

Table of Contents

1 ABOUT US

4 CHAIRPERSON'S ADDRESS

6 EDITOR'S NOTE

7 A MILESTONE TO REMEMBER EVENTS :

8 Sustainable Prototyping with TinkerCad

9 The AI-Driven Future of Electric Vehicles

10 Relay-Thon: AI for Sustainability

11 Powering the Future: Quality & Supply Chain in EV Battery Technology

13 Career Boost: LinkedIn & Resume Building 14 Powering the Future with the Wind!

15 PELS Day 2025

17 AI in Battery Management Systems for Electric Vehicles

GRAVITAS 2025 :

19 ML Using MATLAB 3.0

21 Hackovation 2.0: A 24-Hour Hackathon

23 Outreach Program at Takshilah School

25 OUR FACULTY CO-ORDINATORS

26 ADVISORY BOARD OF 2026-27

30 BOARD OF 2026-27

01

About us

IEEE PELS

IEEE Power Electronics Society

It is more than just a student chapter it is a community of passionate innovators, thinkers, and future engineers united by one goal: to harness the power of electronics for a smarter, cleaner, and more sustainable world

ABOUT US

IEEE — the Institute of Electrical and Electronics Engineers stands as the world's largest professional organization dedicated to advancing technology for the benefit of humanity. With a global community of over 486,000 members spanning 190+ countries, IEEE drives innovation across electrical engineering, computer science, electronics, and beyond. Through its world-renowned publications, landmark conferences, and internationally recognized technical standards, IEEE shapes the future of technology. It publishes nearly a third of the world's technical literature in its domains and has established over 1,000 active standards that underpin the modern digital world from wireless communication to smart energy grids.

IEEE

The IEEE Power Electronics Society (PELS) is one of the fastest-growing technical societies within IEEE, with over 35 years of guiding innovation in power electronics. What makes PELS truly remarkable is where its work shows up inside every electric vehicle, every solar panel, every smartphone charger, and even in spacecraft powering deep-space missions. Power electronics is the invisible backbone of the clean energy revolution, and PELS is the global community driving it forward.

- Championing scientific research and education in power electronics from EV charging and renewable energy to space power applications.
- Keeping members at the forefront through the IEEE Transactions on Power Electronics one of IEEE's most cited journals and global conferences including APEC and ECCE.

pels
IEEE POWER ELECTRONICS SOCIETY
Powering a Sustainable Future

ABOUT US

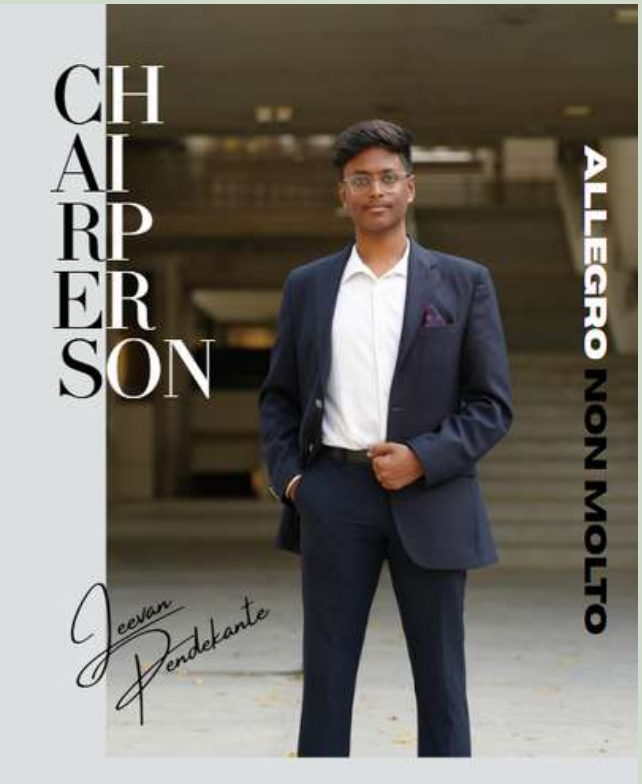
IEEE Power Electronics Society (PELS) VIT Vellore is more than just a student chapter it is a community of passionate innovators, thinkers, and future engineers united by one goal: to harness the power of electronics for a smarter, cleaner, and more sustainable world. Affiliated with the global IEEE Power Electronics Society, the chapter empowers students to explore cutting-edge technologies, engage with industry pioneers, and contribute to research that shapes tomorrow.

Objectives and Activities

IEEE PELS VIT Vellore is committed to transforming curiosity into capability through:
Workshops & Seminars – Immersive, expert-led sessions spanning power converters, renewable energy, electric vehicles, VLSI design, and emerging technologies that are redefining industries.
Technical Projects & Hackathons – High-energy competitions where students step beyond textbooks, collaborate under pressure, and engineer real-world solutions with meaningful impact.
Industry Interaction & Research – Exclusive access to professionals, researchers, and innovators from world-class organizations building bridges between the classroom and the cutting edge.

IEEE PELS VIT Vellore stands as a launchpad for students driven by purpose and fueled by innovation. A testament to this excellence the chapter proudly stands as a 2025 IEEE Regional Exemplary Student Branch Award Winner under IEEE Region 10, earning its place among the most distinguished student branches across the entire Asia-Pacific region.

02 CHAIRPERSON'S ADDRESS



Jeevan P.
Chairperson
IEEE-PELS VIT Vellore

As the Chairperson of the IEEE Power Electronics Society (PELS) VIT Vellore Chapter, I take immense pride in presenting the first edition of our annual newsletter a living testament to the passion, perseverance, and brilliance of every individual who has been part of this journey. This newsletter is not just a publication; it is a reflection of who we are and what we stand for.

Guided by our vision of "Powering Progress through Innovation," we have spent the past year building a strong foundation of learning, research, and hands on experience. From high energy hackathons to thought provoking guest lectures and cutting edge technical workshops, every initiative has been designed to push boundaries and inspire excellence.

Through these efforts, we have not only sharpened technical expertise but also forged meaningful connections with industry leaders, academic researchers, and fellow innovators strengthening the very fabric of our chapter.

Chairperson's Address

Looking ahead, the horizon is filled with exciting possibilities upcoming conferences, collaborative research opportunities, industry partnerships, and events that will continue to elevate our chapter to greater heights. I encourage every member to engage actively, bring fresh ideas to the table, and never stop questioning, exploring, and innovating.

To every student who has been part of this chapter your energy, dedication, and hunger to learn are what make IEEE PELS VIT Vellore truly special. Together, we do not just power circuits we power futures.

heartfelt thank you to our incredible team, dedicated volunteers, inspiring faculty mentors, and supportive industry partners. Your unwavering belief in this chapter makes every achievement possible.

"Together, we power progress—together, we innovate for the future."

Editor's Note

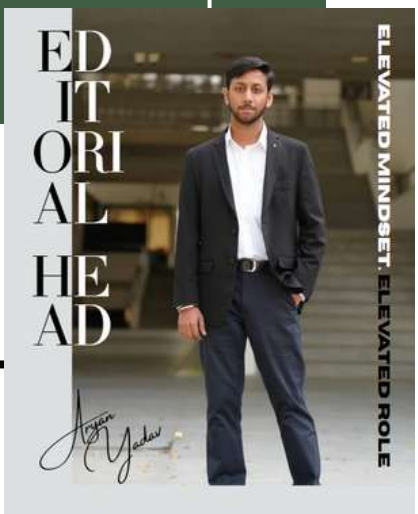
Aryan Yadav

Editorial Head
IEEE-PELS VIT Vellore

03

As we present this edition of our newsletter, it is a moment of both reflection and excitement. The past year has been an extraordinary journey one defined by innovation, collaboration, and a relentless pursuit of excellence that has shaped the IEEE Power Electronics Society Chapter at VIT Vellore into what it proudly stands as today.

Under the inspiring leadership of our former Editorial Head, Sudarshan Ravindran, the chapter reached remarkable new heights. His vision, dedication, and passion for storytelling gave this newsletter its voice turning it from an idea into a publication that truly captures the spirit of our chapter. His contributions have left a lasting mark that continues to guide us forward.



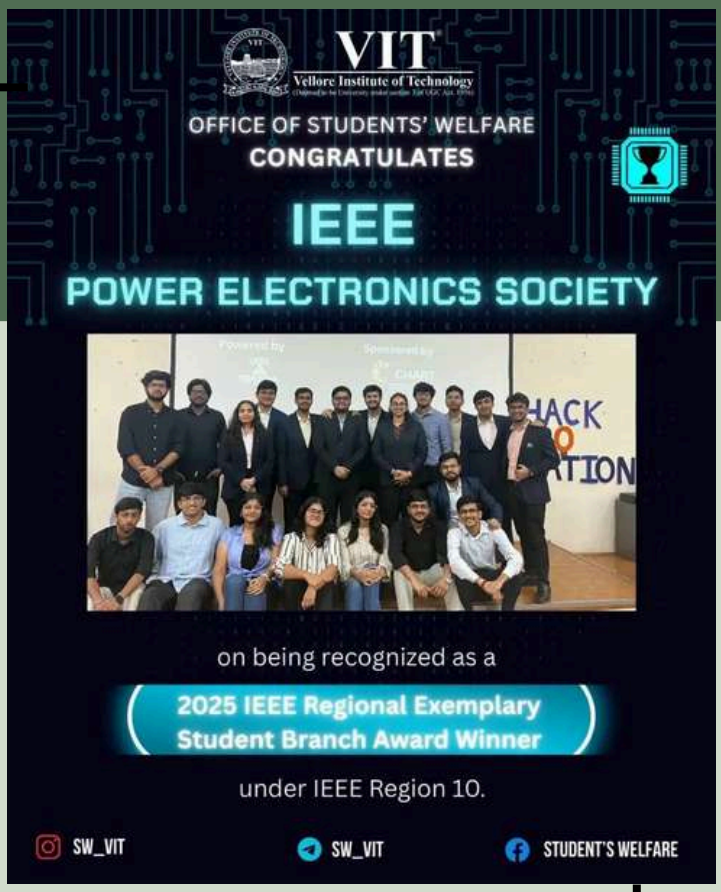
Taking forward this legacy is both an honor and a responsibility I embrace wholeheartedly. Beyond the events, achievements, and milestones documented in these pages, this year has been about the people the friendships forged late at night, the ideas that sparked in workshops, and the shared belief that together, we can power something truly meaningful.

As we look ahead, we do so with deep gratitude for those who built this foundation, and with burning enthusiasm for everything that lies ahead. To every member, contributor, and reader your story is what makes this newsletter worth writing.
Here's to another year of learning, growth, and success!

A MILESTONE TO REMEMBER

04

2025 IEEE Regional Exemplary Student Branch Award Winner | IEEE Region 10



The IEEE Power Electronics Society (PELS) Chapter, VIT Vellore has been honored as the 2025 IEEE Regional Exemplary Student Branch Award Winner under IEEE Region 10 one of the largest and most competitive IEEE regions globally, representing premier institutions across India, Japan, South Korea, Australia, Singapore, and beyond.

This recognition places VIT IEEE PELS among the most distinguished student branches across the entire Asia-Pacific region. As part of this honor, the chapter will be featured live at the Worldwide 2025 IEEE MGA SAC Awards Virtual Ceremony, February 2026.

We extend our heartfelt gratitude to our Faculty Coordinators, Dr. Albert Alexander S and Dr. Chitra A, whose unwavering guidance made this achievement possible.

This is not just an award — it is a reflection of every late night, every event, and every ounce of passion poured into this chapter.

Sustainable Prototyping with TinkerCad

As part of Yantra, VIT's technical week, the School of Electrical Engineering in association with IEEE-PELS organized a hands on workshop titled 'Sustainable Prototyping with TinkerCad' on February 4, 2025. The session introduced students to microprocessors, simulation platforms, and embedded systems for sustainable prototyping. Participants explored ESP8266, TinkerCad, and Wokwi through a simulation of a traffic light control system and servo motor, before advancing to a hardware challenge. The highlight was the development of a weather humidity station a practical, real-world sustainable prototype. Volunteers actively guided participants, ensuring a comprehensive and interactive learning experience that reinforced VIT's commitment to technological innovation

The AI-Driven Future of Electric Vehicles

As part of Yantra, IEEE-PELS organized a tech-talk titled 'The AI-Driven Future of Electric Vehicles' on February 5, 2025, at Rajaji Hall. Dr. Chitra A delivered an in-depth session on how AI is transforming EV technologies covering smart battery management, intelligent transportation, and next-generation mobility solutions. A quiz on AI and Sustainable Development Goals (SDGs) followed, testing participants' knowledge on sustainability innovations.

Prithviraj, Co-Secretary of IEEE-PELS, then led an engaging discussion on AI's broader role in addressing global challenges like climate change, smart agriculture, and energy conservation. Students left with a sharper perspective on the intersection of AI and sustainable mobility.

Relay-Thon: AI for Sustainability

As part of Yantra, IEEE-PELS organized 'Relay-Thon: AI for Sustainability' on February 6, 2025, at Bhagat Singh Gallery. The event was a dynamic AI-driven coding relay competition focused on Sustainable Development Goals. Teams of three tackled coding challenges in a sequential relay format each member building on the previous participant's work without direct communication.

Spanning three rounds, the competition tested adaptability, coding proficiency, and the ability to construct solutions collaboratively under pressure. The event provided a stimulating platform for students to explore AI and sustainability while fostering critical thinking and teamwork.

Powering the Future: Quality & Supply Chain in EV Battery Technology

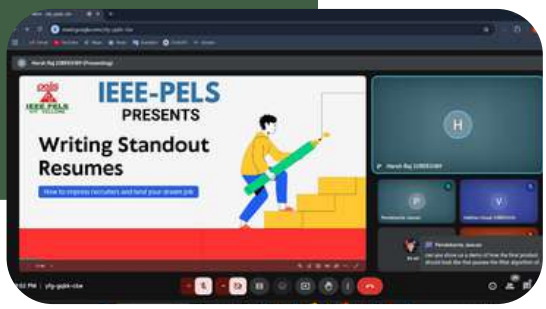
As part of Quality Week at VIT, IEEE-PELS organized an expert lecture titled 'Powering the Future: Quality & Supply Chain in EV Battery Technology' on February 28, 2025, at MB Rajaji Hall. The session was attended by 40 students and faculty members, and was delivered by Dr. Mohan Krishna, Manager Research at TCI-IIMB Supply Chain Sustainability Lab, IIM Bangalore a leading expert on Energy Supply Chain and Hydrogen Transportation.

Dr. Krishna emphasized that high-quality batteries are the backbone of electric vehicles, discussing crucial parameters such as energy density, cycle life, and thermal stability as fundamental benchmarks for superior battery performance. He addressed key supply chain challenges including raw material scarcity, efficiency limitations, safety concerns, and vulnerabilities in supply logistics stressing the importance of strategic material selection, sustainable sourcing, advanced cell designs, smart battery management systems, and effective thermal management.

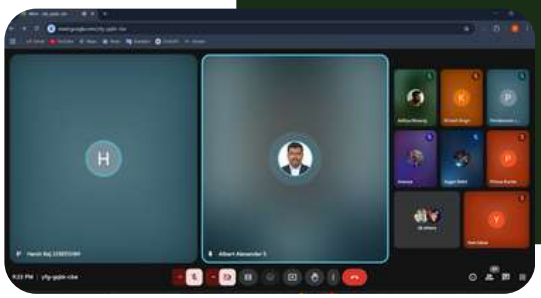
As part of Quality Week at VIT, IEEE-PELS organized an expert lecture titled 'Powering the Future: Quality & Supply Chain in EV Battery Technology' on February 28, 2025, at MB Rajaji Hall. The session was attended by 40 students and faculty members, and was delivered by Dr. Mohan Krishna, Manager Research at TCI-IIMB Supply Chain Sustainability Lab, IIM Bangalore a leading expert on Energy Supply Chain and Hydrogen Transportation.

Career Boost: LinkedIn & Resume Building

On April 13, 2025, IEEE-PELS hosted an online webinar titled 'Boost Career: LinkedIn & Resume Building' via Google Meet. The session was inaugurated by Faculty Coordinator Dr. Albert Alexander S, who emphasized the importance of personal branding in today's competitive job market. Design Head Harsh led two focused sessions the first on LinkedIn optimization, covering profile building, algorithmic visibility, and professional networking;

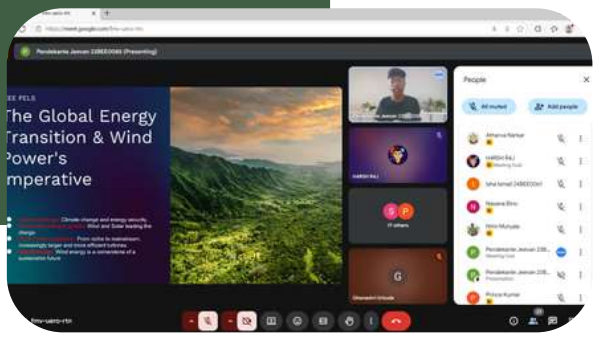


The second on resume crafting, with practical tips on tailoring resumes, formatting, and avoiding generic templates. Students left equipped with actionable strategies to build a strong professional identity and craft compelling, job-ready resumes.

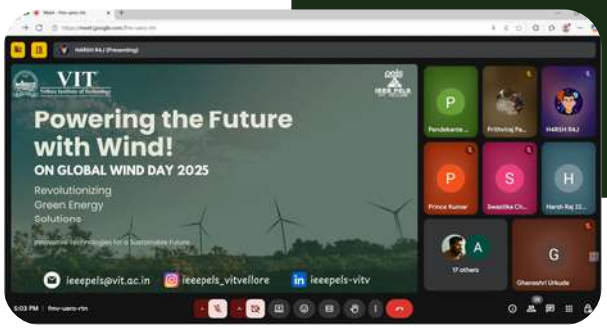


Powering the Future with the Wind!

On June 15, 2025, to commemorate Global Wind Day, IEEE-PELS hosted a webinar titled 'Powering the Future with the Wind!' on Google Meet. Jeevan P. led the core session, covering two themes wind energy generation and its interface with power electronics. He explained how variable-frequency AC power from wind turbines is converted to grid-compatible electricity using AC–DC–AC converters, and discussed modern devices like Silicon Carbide (SiC) and Gallium Nitride (GaN) for efficient performance.

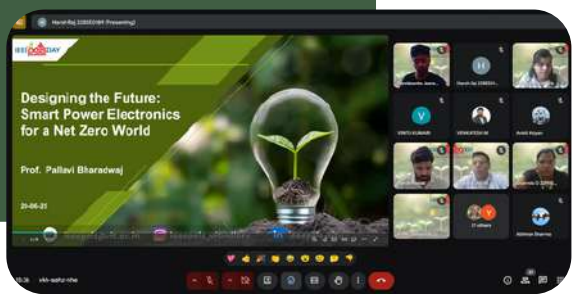


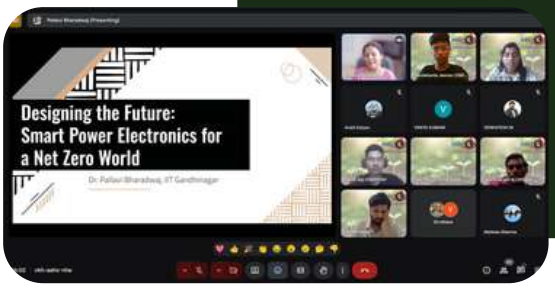
The session also covered MPPT techniques, HVDC systems, and pitch control mechanisms. With over 25 participants, the event successfully highlighted the vital role of power electronics in making renewable energy viable and efficient.



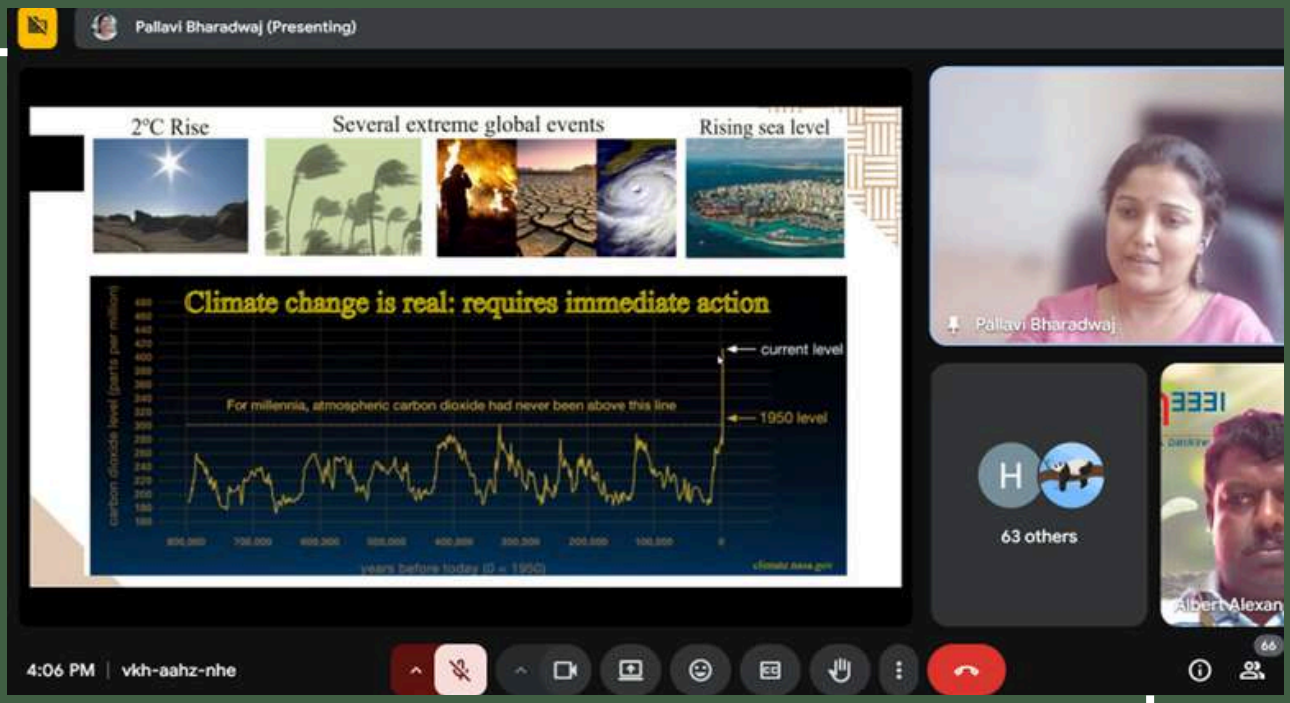
PELS Day 2025

On June 20, 2025, IEEE-PELS celebrated PELS Day with a technical webinar titled 'Designing the Future: Smart Power Electronics for a Net Zero World' on Google Meet.





The keynote was delivered by Prof. Pallavi Bharadwaj from the Department of Electrical Engineering, IIT Gandhinagar. Her talk explored the integration of renewable energy sources using advanced power electronic converters, inverter control strategies, and grid stability mechanisms.



Prof. Bharadwaj's session offered a compelling look into the future of sustainable power systems, highlighting how smart converters and intelligent control algorithms are driving the global transition toward net-zero energy. The event concluded with closing remarks by Jeevan, summarizing key takeaways and expressing gratitude to the speaker, leaving attendees motivated to contribute to the evolving landscape of clean and efficient energy.

AI in Battery Management Systems for Electric Vehicles

On August 4, 2025, IEEE-PELS organized an expert talk titled 'AI in Battery Management Systems for Electric Vehicles' at TT-434A, VIT Vellore, from 10:00 AM to 11:30 AM. The event began with a warm welcome and introduction by M. Varshitha and Nehal Sharma, who set the context by highlighting the growing importance of intelligent energy management in modern electric vehicles. The main session was delivered by Prof. Maher Al Greer, who provided a comprehensive overview of how AI technologies are revolutionizing the design and control of Battery Management System (BMS) architectures.

Prof. Greer elaborated on AI applications in State of Charge (SOC) and State of Health (SOH) estimation, fault detection, and battery lifetime prediction. He also introduced emerging technologies such as Digital Twins and Physics-informed AI for real-time modeling and predictive maintenance, along with the role of big data analytics and edge computing in next-generation EVs.

An engaging interactive Q&A session followed , where participants explored topics ranging from AI career paths and research opportunities to practical tools for innovation in EV technologies. The event concluded with a vote of thanks and a token of appreciation presented to the guest speaker. The session successfully enriched participants' understanding of how AI can drive the next wave of sustainable transportation by improving battery performance, reliability, and overall energy efficiency.

ML Using MATLAB 3.0

On September 26, 2025, IEEE-PELS in collaboration with MathWorks India organized a hands-on workshop titled 'ML Using MATLAB 3.0' at SJT 319, VIT Vellore, from 2:30 PM to 6:30 PM. The session aimed to introduce students to practical applications of machine learning using MATLAB and MATLAB Mobile. The workshop was conducted by Dr. Nisha Chetana Sastry, PhD, from the Education Customer Success Team, MathWorks India, who brings extensive experience in computational neuroscience and applied AI.

Dr. Sastry began with an overview of machine learning fundamentals, covering supervised and unsupervised learning, data preprocessing, model training, and evaluation workflows. She guided participants through MATLAB's ML environment and shared comprehensive resources to help them explore advanced ML tools independently. Participants were then introduced to MATLAB Mobile, learning to sync the application with MATLAB on their systems for seamless real time experimentation.

The highlight of the session was a live demonstration of real time object identification using a mobile camera showcasing how image data can be analyzed through trained ML models, giving attendees a complete understanding of model deployment and real world implementation. The workshop concluded with an interactive Q&A, where students enthusiastically discussed applications of ML across domains like image recognition, healthcare analytics, and automation. The event proved to be highly informative, enabling participants to gain hands-on experience with one of the most powerful platforms for machine learning and data driven innovation

Hackovation 2.0: A 24-Hour Hackathon

IEEE-PELS organized Hackovation 2.0, a 24-hour hackathon, on September 27–28, 2025, at Sarojini Naidu Gallery, VIT Vellore. The event aimed to foster innovation, collaboration, and hands on problem solving among students through a dynamic platform blending creativity with technology. The hackathon commenced at 9:00 AM with a lively introduction by the anchoring club, followed by a sponsor reveal acknowledging the support from valued partners.

Over 24 intense hours, participants engaged in Hack & Build sessions, developing prototypes and applying their technical skills to real world challenges. Periodic mentor review sessions provided constructive feedback, helping teams refine and strengthen their projects through the day and night.

The second day featured a guest session by Mr. Kanishka Mukherjee, CEO and Founder of CHART Techno Solutions, who shared inspiring insights on innovation and technology leadership. His talk motivated participants to think beyond conventional solutions and approach challenges with a creative, entrepreneurial mindset.Shortlisted teams then delivered their final pitches to a panel of judges, showcasing creative solutions and strong technical implementations. Winners were celebrated during the closing ceremony, and the event concluded with a heartfelt vote of thanks. Hackovation 2.0 strengthened the spirit of teamwork and technological excellence within the IEEE-PELS community at VIT.

Event Name	First Place	Second Place	Third Place
HACKOVATION 2.0	Shatter Org	AI-SYBAU	Knights X Ninjas
Prize (INR)	15,000/-	10,000/-	5,000/-

Outreach Program at Takshilah School

IEEE-PELS conducted its Outreach Program at Takshilah School, Vellore, with nearly 70 volunteers from various clubs participating to inspire school students through interactive learning. After a brief orientation on schedules and responsibilities, teams were assigned to different classes, with IEEE-PELS allotted Grade 8 for a one-hour engaging teaching session.

The session covered Active and Passive Voice, a Mathematics Quiz to encourage teamwork and quick thinking, and an introduction to semiconductors through real-life applications. A mind game activity concluded the session, making learning both educational and enjoyable for the students

Students demonstrated remarkable discipline and enthusiasm throughout the program. A short feedback interaction revealed their appreciation for the teaching style and interactive activities. The principal commended the volunteers for their effort and positive impact on the students. The outreach program proved to be a meaningful initiative, strengthening IEEE-PELS's spirit of educational service, teamwork, and community engagement beyond the boundaries of VIT.

Dr. CHITRA A

05

Professor
Grade 1
School of Electrical
Engineering(SELECT) VIT
Vellore

Behind every great chapter stands even greater mentors. Dr. Chitra A and Dr. Albert Alexander S are not just faculty coordinators — they are the guiding stars of IEEE PELS VIT Vellore. Their unwavering belief in the potential of every student, combined with their deep expertise and visionary thinking, has been the cornerstone of everything this chapter has achieved.

From the very first brainstorming session to the grandest of events, their doors have always been open — ready with wisdom, encouragement, and the kind of guidance that turns good ideas into extraordinary realities. It is their relentless dedication that has empowered us to dream bigger, work harder, and reach further than we ever thought possible.

Words will never be enough, but we say it with our whole hearts — thank you. Every achievement this chapter earns carries your guidance within it.

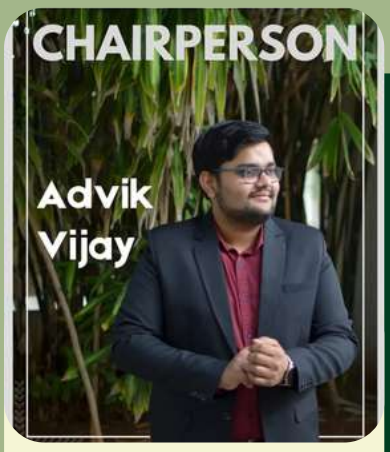
IEEE PELS VIT Vellore is what it is today — because of you.

Associate Professor Senior
Grade 2
School of Electrical
Engineering(SELECT) VIT
Vellore

DR. ALBERT ALEXANDER S

06 Advisory Board of 2026-27

As the torch passes to a new generation of leaders, we celebrate the remarkable individuals who shaped IEEE-PELS VIT Vellore through a year of innovation, dedication, and growth. Their collective efforts have built a stronger, more vibrant chapter — and while their roles may have changed, their guidance and mentorship continue to inspire. Here's a closer look at the Advisory Board of 2026-27.



ADVIK VIJAY

EX-CHAIR PERSON
IEEE-PELS VIT VELLORE

Advik led the chapter with a bold vision and infectious energy, transforming ideas into impactful realities. His leadership fostered a culture of innovation and excellence, elevating IEEE-PELS VIT Vellore to new heights and inspiring every member to push beyond their limits.

Prachi was the organizational backbone of the chapter, keeping every moving part in sync. Her meticulous attention to detail, timely communication, and calm efficiency were the quiet forces behind the smooth running of every event and activity.

PRACHI KUMARI

EX SECRETARY
IEEE-PELS VIT VELLORE

Advisory Board of 2026-27

DIVYAM VIJAYVARGIA
EX VICE CHAIRPERSON
IEEE-PELS VIT VELLORE

Divyam was the steady hand that kept the chapter's operations aligned with its goals. His sharp strategic thinking and ability to bridge gaps between teams ensured that every initiative was executed with precision and purpose.

Prithviraj brought reliability and dedication to his role, supporting the chapter's coordination efforts with consistency. His collaborative spirit and proactive attitude ensured nothing fell through the cracks, making him an indispensable part of the core team.

PRITHVIRAJ PANDHARE
EX Co-Secretary
IEEE-PELS VIT VELLORE

HARSH RAJ
EX DESIGN HEAD
IEEE-PELS VIT VELLORE

Harsh gave IEEE-PELS VIT Vellore its visual voice. Through striking designs, creative posters, and polished presentations, he ensured the chapter's identity stood out and resonated with audiences both on and off campus.

Sudarshan crafted the chapter's narrative with clarity and creativity. Through newsletters, write-ups, and editorial content, he ensured that the chapter's story was told compellingly — preserving achievements and inspiring readers with every publication.

SUDARSHAN RAVINDRAN
EX EDITORIAL HEAD
IEEE-PELS VIT VELLORE

Advisory Board of 2026-27

ANANYA
EX Events Head
IEEE-PELS VIT VELLORE

Ananya was the architect behind the chapter's most memorable experiences. From conceptualizing ideas to flawless on-ground execution, her passion for creating impactful events kept the chapter buzzing with energy and enthusiasm throughout the year.

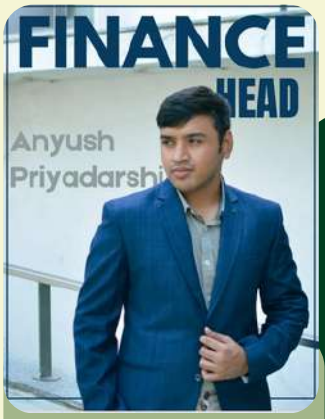
VAIBHAV GOYAL
EX Technical Head
IEEE-PELS VIT VELLORE

Vaibhav was the technical heartbeat of the chapter, driving workshops, projects, and innovations with expertise and enthusiasm. His commitment to hands-on learning empowered members to explore the frontiers of power electronics and emerging technologies.

YASH SAHAI
EX Outreach Head
IEEE-PELS VIT VELLORE

Yash extended the chapter's impact far beyond the campus, building meaningful connections with schools, communities, and organizations. His outreach efforts embodied the chapter's belief that knowledge shared is knowledge multiplied.

Advisory Board of 2026-27

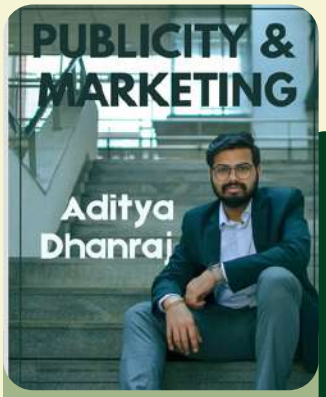


ANYUSH PRIYADARSHI
EX Finance Head
IEEE-PELS VIT VELLORE

Anyush kept the chapter's financial engine running smoothly with careful planning and transparent management. His prudent budgeting and resource allocation ensured that every event was well-funded and executed without compromise.

NITIN MUTYALA
EX Operations Head
IEEE-PELS VIT VELLORE

Nitin was the engine behind the scenes, ensuring every logistical detail was handled with efficiency and care. His operational expertise and problem-solving approach kept the chapter's activities running seamlessly from start to finish.

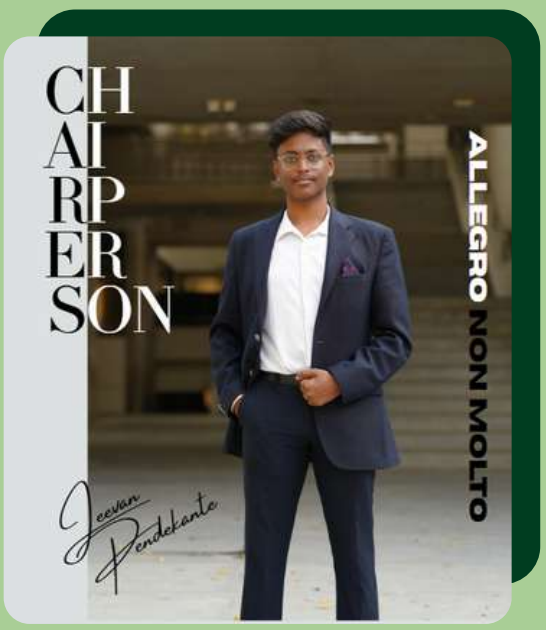


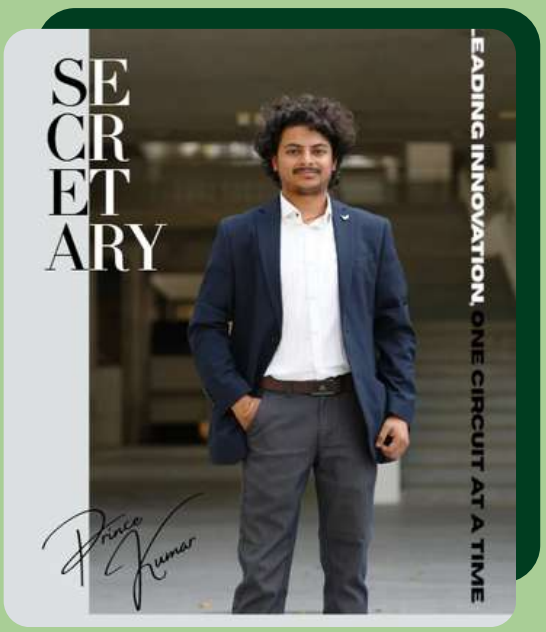
ADITYA DHANRAJ
Ex-Publicity & Marketing Head
IEEE-PELS VIT VELLORE

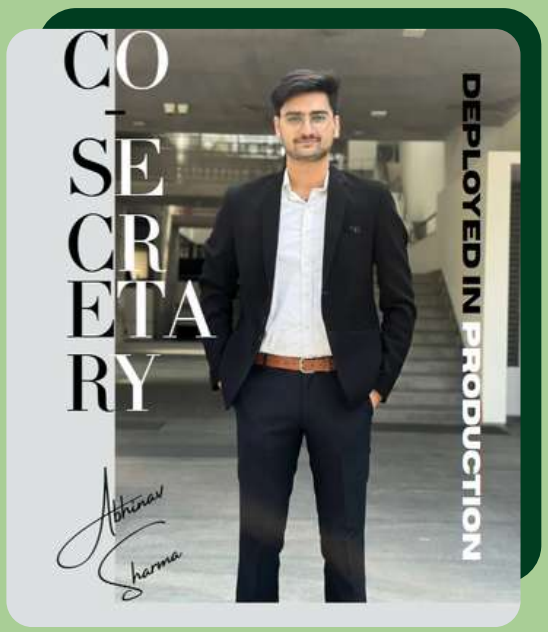
Aditya was the amplifier of the chapter's voice, driving awareness and engagement through smart marketing strategies. His efforts significantly boosted the chapter's visibility, drawing wider participation and building a stronger community presence.

BOARD OF 2026-27

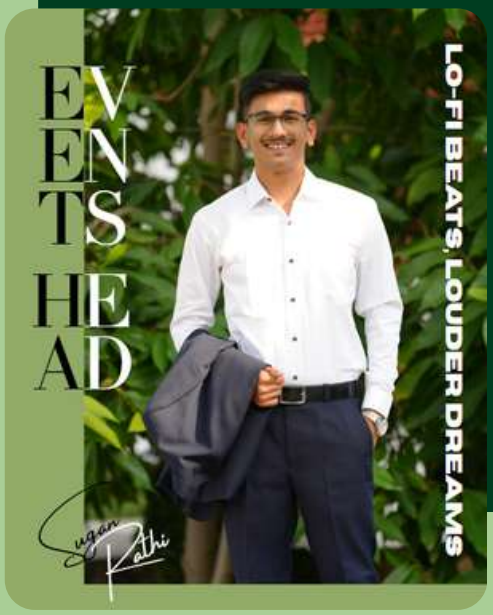
07

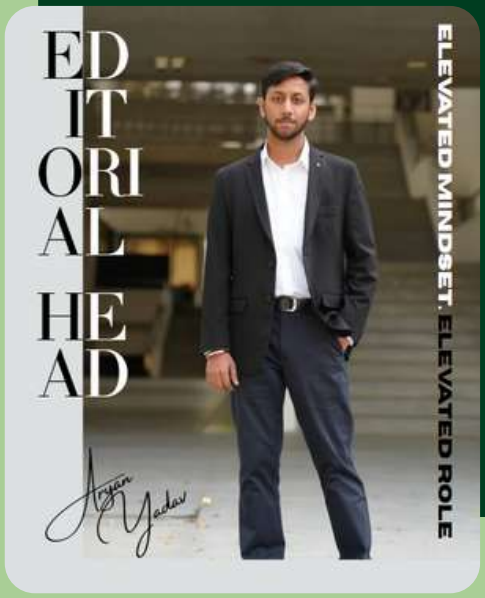






BOARD OF 2026-27





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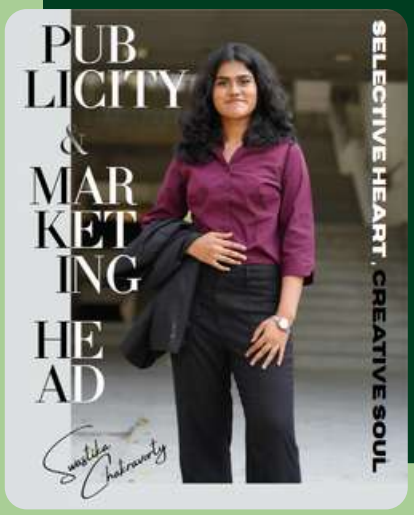

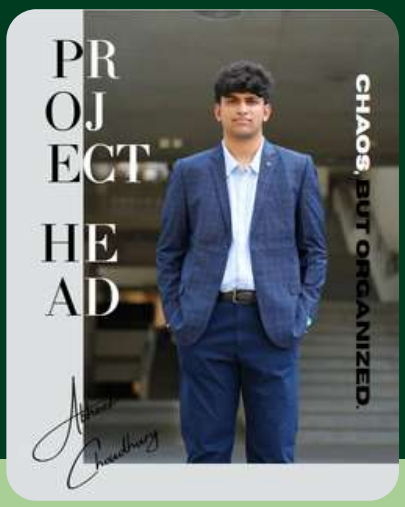

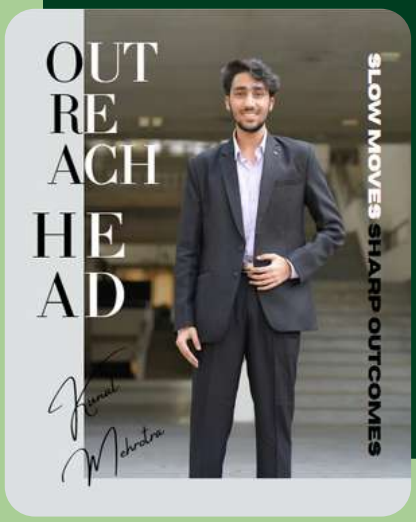

LET'S CONNECT

DESIGN TEAM :

MALAVIKA ANILKUMAR (24BEE0038)
VAISHAKH MANOJ (24BCI0038)

EDITORIAL TEAM:

ARYAN YADAV (24BCT0306)
HARDIK AGARWAL (24BCE2949)